11-9

(a)

$$\begin{matrix} X_{1,1} & X_{1,2} \\ X_{2,1} & \cancel{X_{2,2}} \to 0 \end{matrix}$$

完整性：$X_{2,1}+X_{2,2}=1 \Rightarrow X_{2,1}=1$

互换性：$A_2 X_{2,1} = A_1 X_{1,2}$

$$X_{1,2} = \frac{A_2}{A_1}X_{2,1} = \frac{\pi R^2}{2\pi R^2}X_{2,1} = \frac{1}{2}(1) = \underline{0.5}$$

(b)

$$\begin{matrix} X_{1,1} & X_{1,2} \\ X_{2,1} & \cancel{X_{2,2}} \to 0 \end{matrix}$$

$X_{2,1}=1$

$$X_{1,2} = \frac{A_2}{A_1}X_{2,1} = \frac{\frac{1}{2}\pi R^2}{2\pi R^2}X_{2,1} = \frac{1}{4}(1) = \underline{0.25}$$

(c) 表11-3 圆盘与一个中心在其中垂线上的球

$$\alpha_{1,2} = \frac{1}{2}\left(1-\frac{1}{\sqrt{1+(r_2/h)^2}}\right)$$

无限大平面 $\Rightarrow r_2 \to \infty$

$$X_{1,2} = \frac{1}{2}\left(1-\frac{1}{\infty}\right) = \frac{1}{2}(1) = \underline{0.5}$$

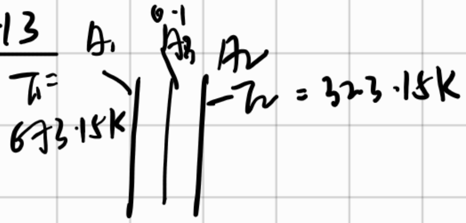
(d)

$$\begin{matrix} X_{1,1} & X_{1,2} \\ X_{2,1} & \cancel{X_{2,2}} \to 0 \end{matrix}$$

$$X_{1,2} = \frac{A_2}{A_1}X_{2,1} = \frac{4\pi R^2}{6(2R)^2}X_{2,1} = \frac{4\pi R^2}{6(4R^2)}(1)$$

$$= \underline{\frac{\pi}{6}}$$

11-13



$\varepsilon_1 = \varepsilon_2 = 0.8$

$A_1 = A_2 = A_3 = A$

板间距远小于板的长度和宽度 $\Rightarrow X_{1,2} = 1 = X_{2,1} = X_{2,3} = X_{1,3}$

$$\Phi_{1,2} = \frac{E_{b1} - E_{b2}}{\frac{1-\varepsilon_1}{A_1\varepsilon_1} + \frac{1}{A_1 \cancel{X_{1,3}}} + \frac{1-\varepsilon_3}{A_3\varepsilon_3} + \frac{1-\varepsilon_3}{A_3\varepsilon_3} + \frac{1}{A_2 \cancel{X_{2,3}}} + \frac{1-\varepsilon_2}{A_2\varepsilon_2}}$$

$E_{b1} = \sigma T_1^4 = 11642.0768 \ W/m^2$

$E_{b2} = \sigma T_2^4 = 618.3006 \ W/m^2$

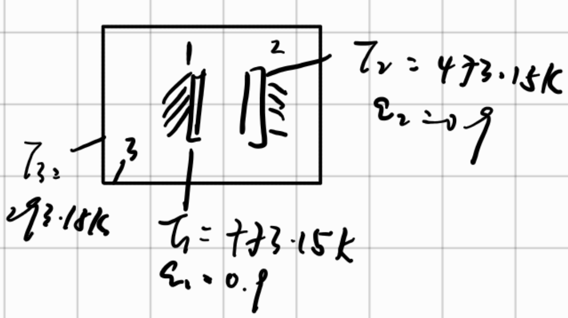
$$\Phi_{1,2} = \frac{A(E_{b1} - E_{b2})}{\frac{1-\varepsilon_1}{\varepsilon_1} + 1 + \frac{1-\varepsilon_3}{\varepsilon_3} + \frac{1-\varepsilon_3}{\varepsilon_3} + 1 + \frac{1-\varepsilon_2}{\varepsilon_2}}$$

$q_{1,2} = \frac{\Phi_{1,2}}{A} = 537.7452 \ W/m^2$

11-14

$x = 1 \quad y = 1$

$$\begin{matrix} \overset{0}{\cancel{X_{1,1}}} & \overset{0.2}{X_{1,2}} & \overset{0.8}{X_{1,3}} \\ \overset{0.2}{X_{2,1}} & \overset{0}{\cancel{X_{2,2}}} & \overset{0.8}{X_{2,3}} \\ X_{3,1} & X_{3,2} & X_{3,3} \end{matrix}$$



PPT中 Figure 13.4

$X_{1,2} = 0.2$ 根据角系数的完整性：$X_{1,1} + X_{1,2} + X_{1,3} = 1 \Rightarrow X_{1,3} = 0.8$

$A_1 X_{1,2} = A_2 X_{2,1} \quad A_1 = A_2 \Rightarrow X_{1,2} = X_{2,1}$

$$\Phi_{1,2} = \frac{E_{b1} - E_{b2}}{\frac{1-\varepsilon_1}{A_1\varepsilon_1} + \frac{1}{A_1 X_{1,2}} + \frac{1-\varepsilon_2}{A_1\varepsilon_2}} = 3483.6472 \ W$$

$E_{b1} = \sigma T_1^4 = 20259.9374 \ W/m^2$ $E_{b3} = \sigma T_3^4 = 418.7383 \ W/m^2$

$E_{b2} = \sigma T_2^4 = 2841.7014 \ W/m^2$

$$\Phi_{1,3} = \frac{E_{b1} - E_{b3}}{\frac{1-\varepsilon_1}{A_1\varepsilon_1} + \frac{1}{A_1X_{1,3}} + \frac{1-\varepsilon_3}{A_3\varepsilon_3}} = 14577.2075 \ W$$

房间可以近似地认为黑体 $\Rightarrow \varepsilon_3 = 1$

$\Phi_1 = \Phi_{1,2} + \Phi_{1,3} = 18060.8547 W$

$\Phi_{2,1} = -\Phi_{1,2}$ $\Phi_{2,3} = \dfrac{E_{b2} - E_{b3}}{\frac{1-\varepsilon_2}{A_2\varepsilon_2} + \frac{1}{A_2X_{2,3}}} = 1733.8504 \ W$

$\Phi_2 = \Phi_{2,1} + \Phi_{2,3} = -1749.7968 W$

11-17

$\alpha = 0.5$ $\varepsilon_1 = 0.2$ $1000 \ W/m^2$ $h = 20 W/m^2 \cdot K$

$$q_{1,2} = \frac{E_{b1} - E_{b2}}{\frac{1-\varepsilon_1}{\varepsilon_1} + \frac{1}{X_{1,2}} + \frac{1-\varepsilon_2}{\varepsilon_2}}$$ $\varepsilon_2 = 1$ $X_{1,2} = 1$

$$q_{1,2} = \frac{\sigma(273.15 + t)^4}{5}$$

$q_{1,3} = h(t - t_f) = 20(t - 35)$

$q_{1,4} = 1000 W/m^2$

$q_{1,2} + q_{1,3} = q_{1,4}$

$$\frac{5.67\times10^{-8}\,(273.15+t)^4}{5} + 20t - 700 = 1000$$

$$t = 52.94^\circ C$$